JMO – 1998

Max Mark – 100 Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all questions.

1.

(a) If n is a perfect square number, what is the next perfect square number?

(b) Find the unit digit of the $1998^{1998} + 1991^{1998} - 1898^{1998}$.

(c) Fill in the blanks: 4% of 4% = ___ %.

(d) In a 100 metre race, Arun finishes 10 metre behind Binoy and 20 metres ahead of Chandan. Assuming their speeds to be constant, determine how many metres ahead of Chandan, Binoy finished?

(e) If the length of a rectangle is increased by 10% and width is decreased by 10%, then find the % of increase or decrease in area.

(f) Delete 10 digits from the following number 12345123451234512345, so that the new number formed with the remaining digits is the largest possible

(g) The GCD of two numbers is 25 and LCM of the same two numbers is 1400. If one of the two numbers is 175, find the second number.

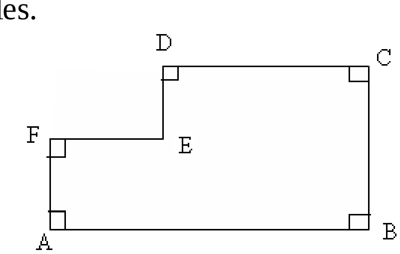
(h) If a, b, c are three positive integers such that a.b.c = 1729, then find the value of a + b + c.

(i) Find the whether the number 2491 is a prime or a composite number?

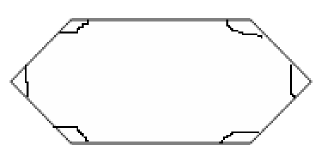
(j) Find largest of the following numbers: $2^{2^{2^2}}, \left(\left(2^2\right)^2\right)^2, 2^{2\times2\times2}$.

2.

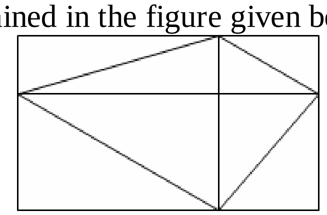
(a) Determine the perimeter of the given figure if AB = 18 cm, BC = 8cm, and angles at the vertices, A, B, C, D, E & F are right angles.



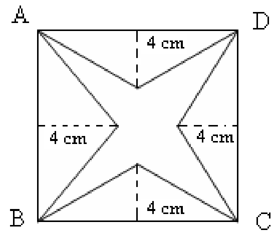
(b) Find the sum of measures of the interior angles of the hexagon shown below.



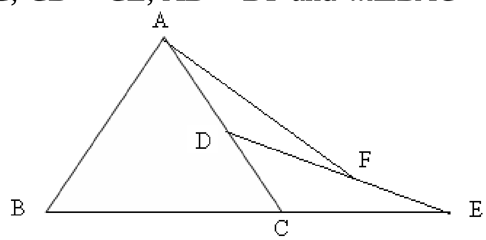
(c) Find the number of triangles contained in the figure given below.



(d) If in the figure below, ABCD is a square of length 12 cm and the height of each triangle is 4 cm (as shown in the figure), then find the area of the star.

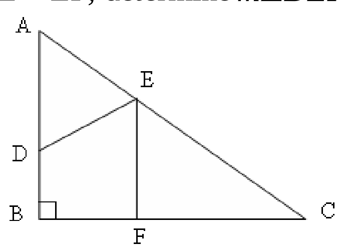


(e) In the figure below, AB = AC, CD = CE, AD = DF and $m\angle BAC = 40^0$. Find $m\angle AFE$.

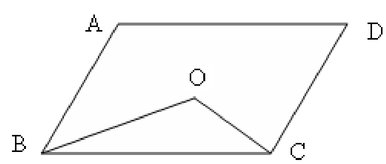


(f) The diagonals of a quadrilateral intersect at right angles and are of lengths 7.6 cm and 5.6 cm. Determine the areas of the quadrilateral.

(g) In the figure below, $m\angle B$ of triangle ABC is right angle. D, E, F are points on AB, Ac and BC respectively such that AD = AE, CE = EF, determine $m\angle DEF$.



(h) In the figure below, ABCD is a parallelogram. O is an interior point of parallelogram such that $m\angle BOC = \frac{1}{3} m\angle ABC$ and $m\angle OCB = \frac{1}{3} m\angle DCB$. Determine $m\angle BOC$.



3.

(a) Show that in a diagram how to plant 10 trees in five lines so that in each line there are four trees.

(b) Determine all the primes between 144 and 200.

4.

(a) After completing a journey by a car, the driver records the distance traveled from the speedometer and calculates the speed taking the time he took to complete the journey from his watch. The speed thus determined was 38 km/h. If the speedometer shows 95 km, when the car travels 100 km and if his watch gains 4 minutes every hour, determine the actual speed with which he drove his car.

(b) How many numbers are there between 1 and 600 (both inclusive) that are neither divisible by 5 nor divisible by 7?

5.
- (a) Out of 5 questions in a paper 5% of the candidates answered all and 5% answered none. Of the rest, 25% answered only one question and 20% answered four questions. If $24\frac{1}{2}\%$ of the total candidates answered only two questions and 200 candidates answered only 3 questions, find out the total number of candidates.
- (b) The sum and product of two mutually prime numbers are 101 and 2520 respectively, determine the numbers.
- (c) A shopkeeper bought 50 school bags for a total cost of Rs. 480 and marked Rs. 16 on each of them, but during sale he allowed a discount of 10%. If 4 bags were damaged and find unfit for sale, determine % of profit or loss on the sale of the rest.

6.
- (a) Out of 27 similar looking coins, one is fake coin whose weight is less than that of genuine coin. If a beam balance is given to you. Chalk out a strategy to detect the fake coin by using balance exactly thrice.
- (b) On the sides and interior of a plot of land in shape of an equilateral triangle with its sides measuring 32 metres each, coconut plantation is to be so made that the distance of each plant from the nearest one is 4 metres. The labour charge for planting 4 or a part thereof is Rs. 10. Determine the total cost of coconut plantation in the land.

7.
- (a) A train, an hour starting from a station, meets with an accident, which detain it for half an hour. Then it proceeds with a speed of ¾ th its former speed and arrives the next station 3½ hour late. Had the accident occurred at 90 km farther from the former place of accident along the line, it would have been 3 hour late. Find the distance between the two stations.
- (b) The price of 3 pens is Rs. 6 more than the price of 5 pencils. If the total price of 14 pencils and 12 pens is Rs. 126, determine the price of one pen and pencil separately.

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